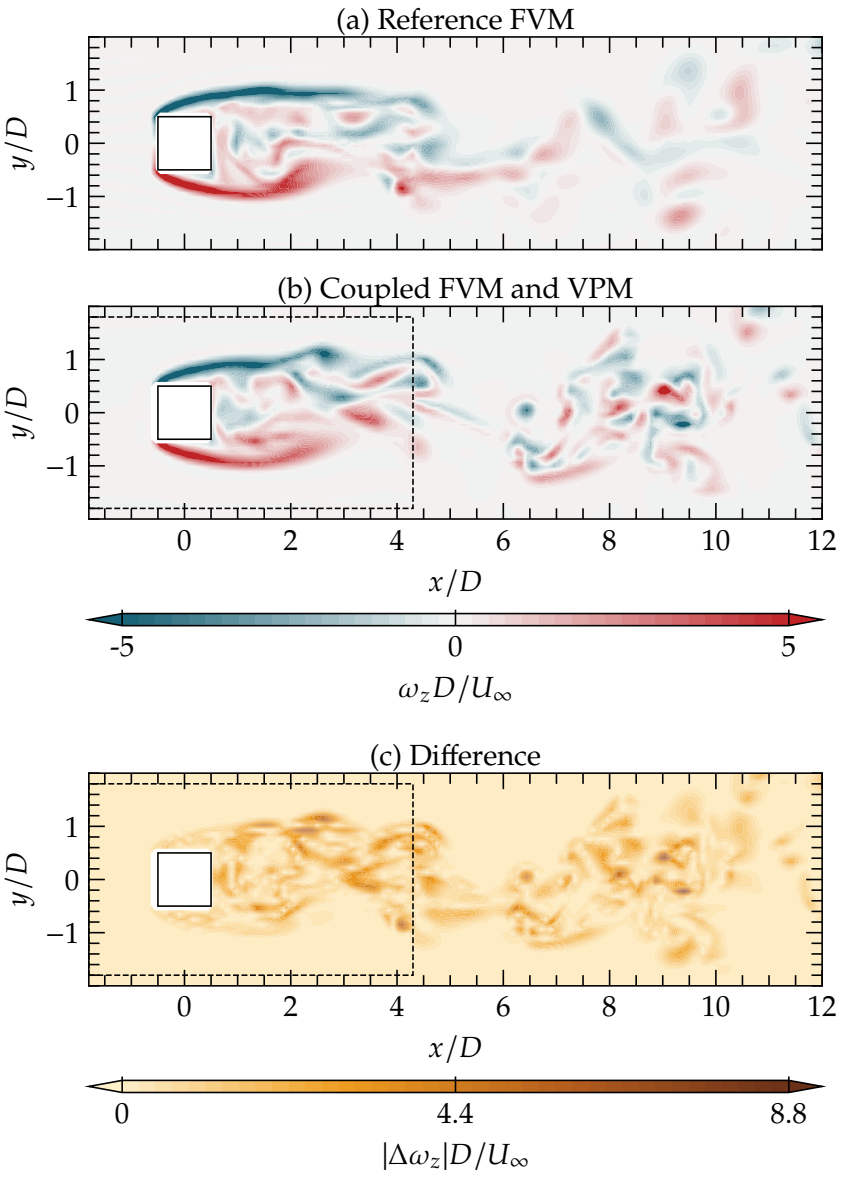

(a) Reference FVM
y/D
1
0
−1
(b) Coupled FVM and VPM
y/D
1
0
−1
0
2
4
6
8
10
12
x/D
-5
0
5
ω_z D/U_∞
(c) Difference
y/D
1
0
−1
0
2
4
6
8
10
12
x/D
0
4.4
8.8
|Δω_z|D/U_∞